

HARSHAN A M

Full Stack AI Engineer — On-Prem & Agentic AI, RAG, NLP, Voice AI, Real-Time & IoT Systems

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SUMMARY

Full Stack AI Engineer with 5.5 years shipping AI-powered, real-time, and scalable systems across EdTech, Industrial IoT, AgriTech, and enterprise tooling for aerospace manufacturing. Depth in LLM systems and RAG (naive to agentic), on-premise and air-gapped agentic AI (LangGraph, PydanticAI, MCP, local LLMs), NLP, Voice AI (ASR/TTS), and data-intensive architectures (Kafka, TimescaleDB, time-series). Strong end-to-end ownership, from UX/UI and system design through production, in JavaScript, TypeScript, and Python.

EXPERIENCE

Software Engineer — Full Stack & AI Systems — Jan 2023 – Present

Lingotran, Rangsons Group — Mysore, India

Core platform — EdTech AI

- Built LLM-powered AI tutor and recommendation systems (OpenAI, Gemini, Ollama) serving learners across multiple language tracks.
- Designed production RAG pipelines (LangChain, LangGraph, Chroma) using HyDE, RAPTOR, and reranking (BGE-M3, Cohere) to lift retrieval precision over naive baselines.
- Built real-time conversational AI on LiveKit and WebSockets, with ASR/TTS via Whisper, Deepgram, and Kokoro TTS, and pronunciation scoring (Soundex, Levenshtein).
- Automated AI-generated learner performance reports and built gamified learning modules with real-time feedback; added dynamic CMS audio via ElevenLabs.
- Instrumented LangFuse observability (prompt tracing, cost, evaluation), built MCP server integrations, and delivered NLP (SpaCy) across Portuguese, Spanish, and French.
- Owned delivery as Scrum Leader; trained interns and contributed to hiring.

Selected platforms delivered across the group. Proprietary work; no public repositories.

Graylinx — OMNYX and THERMYNX (Industrial IoT, On-Premise Agentic AI)

- Architected OMNYX, an on-premise air-gapped Industrial IoT platform: BACnet to Kafka to a dual database core (PostgreSQL 16 and TimescaleDB) with zero cloud data egress.
- Owned the agentic AI layer: Planner, Executor, and Validator agents on a local LLM (Ollama, vLLM) with pgvector RAG and Keycloak-modelled agent authorization.
- Built THERMYNX, the shipping HVAC vertical (FastAPI and React), turning chiller telemetry into explainable insight via 5 autonomous tool-calling agents with SSE-streamed reasoning traces.
- Architected Digital-Twin fault detection and Remaining-Useful-Life prediction, plus reinforcement-learning efficiency optimization with a safe shadow-to-live rollout.
- Stood up the on-premise stack (Docker, Keycloak, Grafana, Prometheus) and stress-tested the Kafka pipeline toward approximately 100x scale (around 1,100 controllers) to size hardware.
- Designed the analytics data warehouse and metadata-driven Airflow ETL, and authored the platform HLD and LLD specifications.

Nesso — Farm-to-Fork Traceability Platform (AgriTech)

- Architected an offline-first traceability platform on a TypeScript monorepo (NestJS 10, Next.js 15, Expo, MongoDB, Redis) with SQLite-outbox offline sync and GPS polygon farm mapping.
- Delivered public QR traceability, 12-language internationalization, WCAG 2.2 AA accessibility, and 17-role RBAC across mobile, web, and QR portal (approximately 99 screens).

Rangsons Aerospace (RAPL) — Enterprise Tools (Full-Stack, Front-End, UI/UX)

- Built a 7-role Customer RFQ to Quotation platform with gated approvals (12 steps, 8 states, 4 gates), revision history, and ECharts reporting, on React 18, Node.js, Express, and MongoDB.

- Built a Manufacturing Execution System handling shift and task tracking, dynamic sequence enforcement, and lockout and orphan-task concurrency controls.
- Owned front-end and UI/UX: a shared design system (light and dark) reused across RFQ, MES, and an IT Helpdesk console (kanban board with SLA indicators).
- Delivered an on-site AI hardware suitability evaluation.

IRIS — B2B Connected Fragrance-Diffuser Platform (IoT)

- Delivered a BLE fleet-operations platform: React 19 admin and Expo React Native field app communicating with hardware over Bluetooth, on Express, Socket.IO, and MongoDB on Azure, covering app development, multi-generation BLE device integration, and UI/UX.
- Built the core loop: onboarding, commissioning, consumption-ledger dry-out prediction, and refill dispatch, with offline-capable field calculations.
- Authored the v1 to v2 system redesign: stateless Redis-backed Socket.IO, a BullMQ worker tier, an MQTT to time-series telemetry pipeline, Key Vault, OpenTelemetry, CI/CD, and offline-first sync, taking readiness from pilot to enterprise-grade.

Neural-Nexus — Hybrid RAG Knowledge Engine (Health)

- Built a hybrid RAG engine over Neo4j (graph) and PostgreSQL (relational) with async FastAPI, parallel multi-source retrieval, and custom merge and rank scoring.
- Enforced strict grounding to minimize hallucinations, with SSE streaming for progressive delivery and Redis/Valkey caching for high-frequency graph queries.

Examic — AI Code Generation Platform

- Built CodeMint: natural language and PRD to code to GitHub to Vercel/Azure deployment automation; integrated OpenHands agent workflows; contributed to team formation and hiring.

Freelance Software Developer — Aug 2022 – Dec 2022

- Delivered full-stack applications, dashboards, and legacy-system improvements for clients end to end, bridging directly into the Lingotran role.

Software Engineer — Full Stack — Feb 2021 – Aug 2022

Veriteam Software Solutions

- Delivered 40+ production applications across CMS, CRM, and e-commerce using Java (Spring, Hibernate), JSP, JavaScript, and Python.
- Integrated Stripe, Razorpay, OAuth, JWT, and RBAC; modernized legacy PHP and .NET codebases.
- Mentored 250+ students and interns across full-stack development.

PROJECTS

- **Nexus Swarm** (github.com/Kimosabey/nexus-swarm) — Stateful multi-agent research and synthesis engine on LangGraph using a Manager-Worker pattern (Orchestrator, Researcher, Writer, Reviewer), with human-in-the-loop breakpoints and PostgreSQL-checkpointed recovery. FastAPI and Next.js 14, Pydantic V2 typed state.
- **Grounded Voice Assistant** (github.com/Kimosabey/kimo-nexus; live demo at kimo-nexus.vercel.app) — Retrieval-grounded assistant with voice input/output and page-level tool-calling. NDJSON token streaming from a Next.js route handler, browser-native speech-to-text and text-to-speech via the Web Speech API (no vendor, no per-request cost), per-IP token bucket with a daily budget guard, 3-model provider fallback so a retired model cannot take it down, and a deterministic no-LLM retrieval path so it degrades instead of failing. Answers are constrained to a fixed knowledge base to prevent fabricated claims.
- **DocMind AI** (github.com/Kimosabey/docmind-ai) — Hybrid RAG (dense and sparse) over uploaded documents using FastAPI, ChromaDB, and Llama 3, with multi-document ingestion, semantic chunking, and query-time reranking.
- **LiveNexus AI** (github.com/Kimosabey/live-nexus-ai) — Real-time speech intelligence: WebRTC to LiveKit SFU to a Faster-Whisper worker with WebRTCvad silence gating and adaptive model selection under CPU load, with a double-buffered live transcription interface.
- **InferenceHub** (github.com/Kimosabey/inference-hub) — Polyglot inference gateway: Node.js for I/O, Zod validation, and auth, with Python for PyTorch and Scikit-learn inference over gRPC and Protocol Buffers using shared .proto contracts, fully containerized.

- **VoiceSync AI** (github.com/Kimosabey/voicesync-ai) — Privacy-first offline transcription: React to pre-signed MinIO upload to FastAPI webhook to Faster-Whisper, fully air-gapped.

SKILLS

AI, GenAI, and NLP: LLMs (OpenAI, Claude, Gemini, Ollama, vLLM for local and on-premise), RAG (naive to agentic), LangChain, LangGraph, PydanticAI, CrewAI, AutoGen, HyDE, RAPTOR, reranking (BGE-M3), MCP server development, tool-calling agents, RAGAS, SpaCy, HuggingFace

Voice AI: Whisper, Faster-Whisper, Deepgram, ElevenLabs, Kokoro TTS, WebRTCvad, LiveKit, SRT streaming, Web Speech API

Languages: JavaScript, TypeScript, Python, SQL (exposure: Java, Golang)

Frontend: React, Next.js 15, TailwindCSS, shadcn/ui, Radix, Framer Motion, Zustand, React Query, TanStack Table, ECharts (exposure: React Native, Expo, Three.js)

Backend and APIs: Node.js (Express, NestJS), FastAPI, Django, REST, GraphQL, gRPC, microservices, modular monolith

Data and Streaming: PostgreSQL, TimescaleDB, MongoDB, Redis, Valkey, Neo4j, ClickHouse, MySQL, Kafka, Airflow, WebSockets, SSE, Protocol Buffers, BACnet, MQTT, metadata-driven ETL, offline-first sync

Vector Stores: Chroma and pgvector in production; Qdrant, Weaviate, and Pinecone familiar

DevOps and Cloud: Docker, CI/CD (GitHub Actions), Turborepo, pnpm, AWS, Azure, Vercel, Kubernetes, Linux

Observability and Auth: Grafana, Prometheus, OpenTelemetry, LangFuse, Keycloak, JWT, OAuth, RBAC

Core Concepts: System design, distributed systems, RAG architecture, LLM evaluation, accessibility (WCAG 2.2 AA)

Domains: Industrial IoT and building management, manufacturing execution (MES), AgriTech traceability, EdTech, voice and real-time systems

EDUCATION

Master of Computer Applications (MCA) — NIE Institute of Technology, Mysore, 2017–2020, 8.2 CGPA

Bachelor of Computer Applications (BCA) — Vidya Vikas College, University of Mysore, 2014–2017, 71%

LEADERSHIP AND ACTIVITIES

- Tech Lead and Organizer for a 24-hour hackathon at MIT College, Mysore, including problem design and mentoring.
- Trained interns across full-stack and AI/ML development; led developer hiring for the Examic team.